

Party planning & booking system

CS 1151_Fundamentals of Object Oriented Programming project

Deena Alazzeh 24-0770

Maryam Alzadjali 24-0566

Atika Al-Hinai 24-0154

Teeb Alharthy 24-0153

Reem Al-Seyabi 24-0417



Features:



Description:

A system used by party planners to make client booking more convenient, displaying all services offered transparently by displaying the service details, duration, and prices. It helps party planners keep track of all their clients, their needs, and any specific details they require.

Party type

Blrthday, Retirement, promotion, Baby shower, Gender reveal, Bachelorette/bachelor party

Venue

Home, park, Banquet ballroom, Rooftop garden, Hotel conference room, Restaurant

Services

Decoration packages (Standard, Luxury, Premium) & Catering options (Basic, Premium, VIP)

Additional fetures

Photography, Videography, Music, Entertainment & Kids Activities

class & responsibility

Class client:

This is a derived class from the parent class its responsibility is related to the users specific data such as payment methods ,booking history.
Service providers

Class event :

This class coordinates the details , its job is to link a specific client to a specific venue and list the services.

Class venue:

iclass that specifies the physical location where the event happens it stores details about the venue. Such as (location, capacity , rental price ,indoor or outdoor)it also checks the availability of the venue at that specific date.

Class booking:

This class calculates the final bill .



class & responsibility

Class service:

it's a base class of all services the company provides it holds the basic items that every class has (Name, price).

Class catering:

it is a class that inherits from the base class (service) which holds anything related to food and beverages that the company can provide It includes a menu and guest count .

Class decoration:

It inherits from the base class (service), manages the theme of the event .

Class entertainment:

It inherits from the base class(service), handles the scheduling and requirements for performers, DJs, or guest speakers.

Class photography:

It inherits from the base class (service), manages the media requirement , number of photographers ,hours of coverage.

1

Client

- name
- email
- number

2

event

- Birthday
 - grad
 - gender
- } Type

- date + time

- number of people (using switch)

1 50 - 100

2 100 - 150

3 150 - 200

4

catering

depend on
number of people

if 50 - 100 ~ 100 OMR

or 100 - 150 ~ 130 OMR

or 150 - 200 ~ 150 OMR

photography

50 pic ~ 20 OMR

100 pic ~ 30 OMR

300 pic ~ 50 OMR

Service

- price
- availability
- Service ID

Entertainment

- DG ~ 50 OMR
- Firework ~ 150 OMR

Decoration

- classic ~
 - Modern ~
- } Switch method
for both according to
number of people

(display,

5

Booking

- cost

Has-A:

- Venue
- Service
- client
- event

3

Venue

- park
- Home
- Hall

OOP principles Demonstrated

OOP Principle	Implementation
Encapsulation	All attributes are private/protected with public getters/setters
Inheritance	Client inherits from Person; Catering/Decoration/Entertainment/Photography inherit from Service
Polymorphism	Virtual functions displayInfo() and displayService() with overrides in each subclass
Composition	Booking class contains Client, Event, Venue, and Service objects (HAS-A relationships)
Memory Management	Proper use of constructors/destructors with new/delete for dynamic allocation